



AFTER

After the war, the combination of scientific discoveries and practical battlefield experience set the stage for greater survival rates in future wars.

PIONEERS OF MEDICINE

Medical practitioners during the Civil War could only do so much. Bad sanitation often got the better of them, and infections were difficult to combat. **But the world of medicine was changing.** In France during the 1860s, the chemist and microbiologist **Louis Pasteur** proved his **germ theory**—that bacteria from the environment caused disease—and discovered vaccines against rabies and anthrax. In 1865, Englishman **Joseph Lister used carbolic acid** to prevent infection in the wound of an 11-year-old boy with a compound fracture of his tibia.

ANTIBIOTICS DISCOVERED

Alexander Fleming's accidental **discovery of the antibiotic penicillin** in 1928 changed the course of medical practice forever. In **World War II, thousands of soldiers were saved** from disease and postoperative complications.